

CASE 20

HYPER IgE SYNDROME

A defect in T_H17 lineage differentiation.

After antigen recognition and concomitant activation by specific sets of cytokines, naive $CD4^+$ T cells differentiate into distinct lineages of helper or regulatory effector T cells, characterized by distinct biologic functions and cytokine expression profiles. About ten years ago, T_H17 cells were identified as a distinct lineage of differentiated $CD4$ T cells, characterized by expression of the cytokines IL-17A (IL-17), IL-17F, IL-21, and, in some cases, IL-22 (Fig. 20.1). IL-17 and IL-17F have similar activities that include the induction in epithelial cells of both chemokines that attract neutrophils and β -defensins (antimicrobial peptides). T_H17 cells play a role in the immune response to extracellular bacteria and fungi, and in autoimmune diseases.

In humans, IL-6, in synergy with IL-1 β , IL-21, and possibly TGF- β , has a key role in the differentiation of T_H17 cells. IL-23 is important in amplifying the generation of T_H17 cells and sustaining their survival. IL-6, IL-21, and IL-23 all bind to the same class of heterodimeric receptors, which, upon binding, activate Janus kinases (JAKs) that phosphorylate the intracellular portions of the receptor. The inactive form of the transcription factor STAT3 is recruited to the phosphorylated receptor and is itself phosphorylated (Fig. 20.2). Phosphorylated STAT3 dissociates from the receptor and forms homodimers that translocate to the nucleus and act as transcription factors to induce expression of the retinoic acid-related orphan receptor γ (ROR γ T). The differentiation of activated $CD4^+$ T cells into T_H17 cells is determined by the expression of the transcription factor ROR γ T.

This case was prepared by Raif Geha, MD, in collaboration with Lisa Bartnikas, MD.

TOPICS BEARING ON THIS CASE:

Neutrophil function

T-cell subsets and their differentiation

T_H17 cell function

IgE class antibody

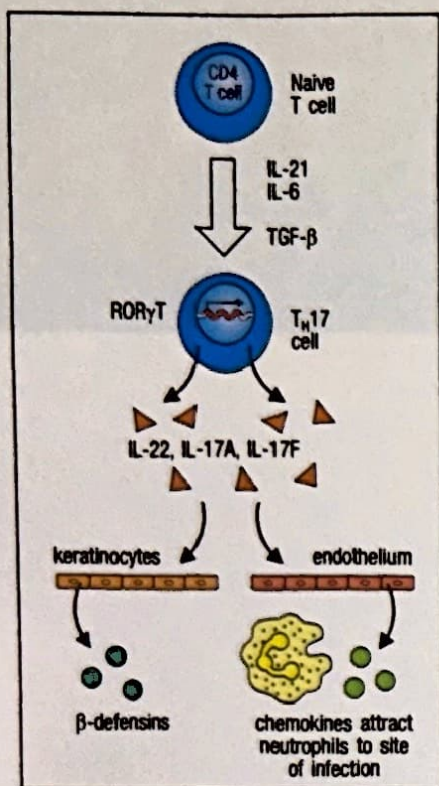


Fig. 20.1 Differentiation and actions of T_H17 cells. Effector T_H17 cells develop from antigen-activated naive CD4 T cells in the presence of the cytokines IL-6, IL-21, and TGF- β . They are characterized by expression of the transcription factor ROR γ T and produce the cytokines IL-17A (IL-17), IL-17F, and IL-22. The cytokines produced by T_H17 cells act on skin to induce the production of antimicrobial defensins, and on endothelial cells to induce the production of chemokines that recruit neutrophils to sites of infection.

The case of Stephen Dedalus: impaired T_H17 differentiation associated with elevated IgE and recurrent infections.

Stephen Dedalus was first taken to the Children's Hospital at 3 years of age for evaluation of his eczema. Stephen was born after an uncomplicated pregnancy, but within the first few days of life he developed significant seborrheic dermatitis (a flaky, red skin rash commonly affecting the scalp) and neonatal acne. By the first month of life, he developed eczema that was at times severe. Aggressive topical skin care was recommended. His parents were both healthy.

Over the next few years, Stephen experienced recurrent infections, but typically was afebrile (did not develop a fever) during these episodes. He had recurrent skin infections, with cultures growing *Staphylococcus aureus*. Although he developed pustules, they were not particularly red or tender. He also had an episode of *S. aureus* bursitis (inflammation of the sac that contains joint fluid) of the knee. He had two episodes of pneumonia, one being complicated by a small pneumothorax (collapsed lung due

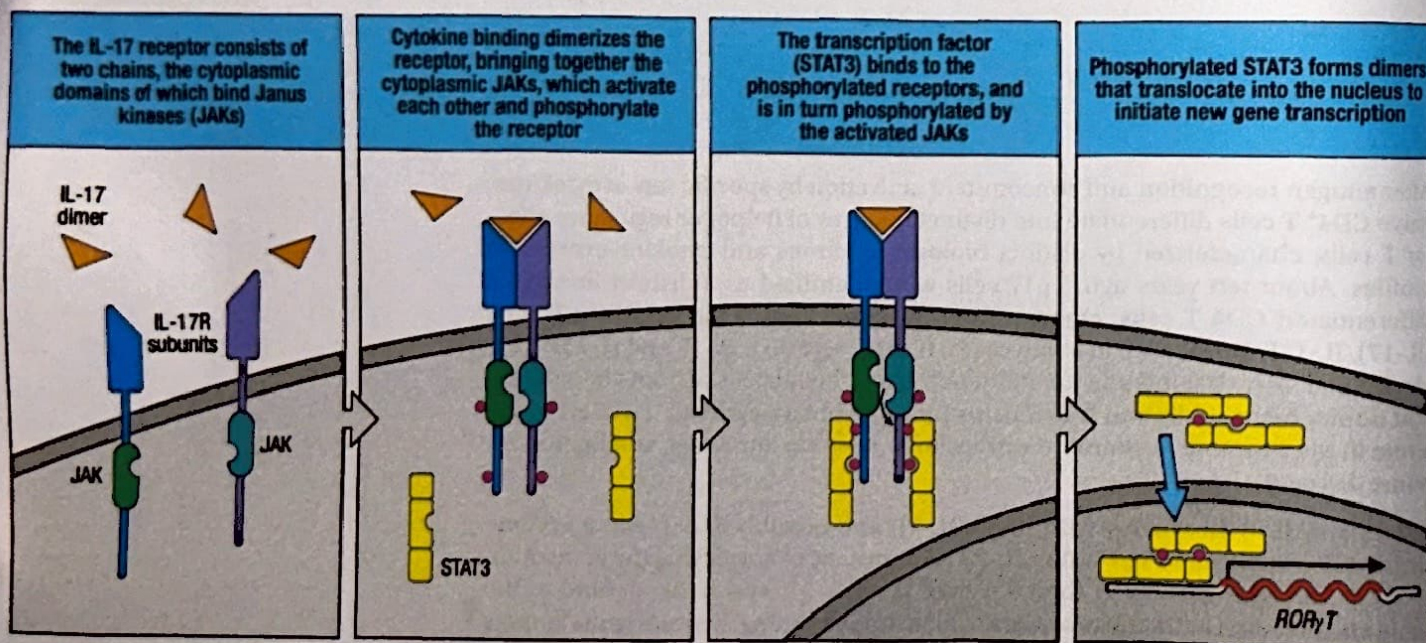


Fig. 20.2 STAT3 is involved in signaling pathways from some cytokine receptors. Many cytokines act via receptors that are associated with cytoplasmic Janus kinases (JAKs). The receptor consists of at least two chains, each associated with a specific JAK (first panel). Binding of dimeric ligand results in dimerization of the receptor chains, bringing together the JAKs, which can phosphorylate and activate each other. The activated JAKs then phosphorylate tyrosines in the receptor tails (second panel). Members of the STAT (signal transducers and activators of transcription) family of proteins (which contain SH2 domains)

bind to the tyrosine-phosphorylated receptors and are themselves phosphorylated by the JAKs (third panel). After phosphorylation, STAT proteins form a dimer due to the SH2 domains binding phosphotyrosine residues on the other STAT, and translocate to the nucleus (last panel), where they bind to and activate the transcription of a variety of genes important for adaptive immunity. Different STATs are associated with different receptors. STAT3, for example, is activated by the receptors for the cytokines IL-6 and IL-10, among others.

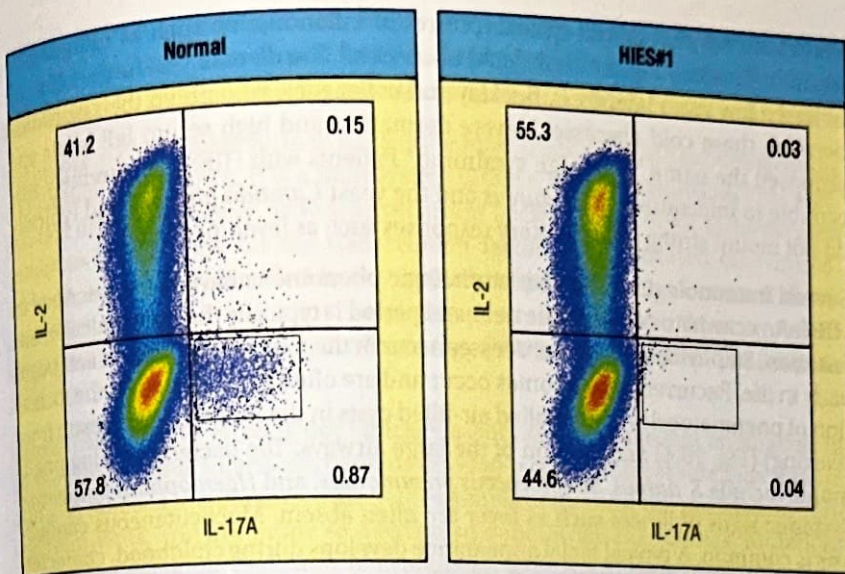


Fig. 20.3 T cells from a patient with hyper IgE syndrome due to a mutation in STAT3 fail to differentiate into T_H17 cells *in vitro*. T_H17 cells (boxed) are evident in the normal sample but absent from HIES sample. Photograph courtesy of Cindy Ma.

to air leakage from the lung into the chest cavity). He had 10 episodes of rhinosinusitis (nose and sinus infection) and a chronic rhinorrhea (nasal discharge) that was typically yellow or green. He had recurrent episodes of otitis media (middle-ear infection) and needed tympanostomy tubes inserted on multiple occasions. These are ventilation tubes inserted into the eardrum to relieve obstruction of drainage from the middle ear to the nose resulting from scarring due to frequent middle-ear infections. Stephen fractured his right arm on two occasions following a minor trauma, suggesting increased susceptibility to bone fractures.

Stephen was admitted to Children's Hospital at 5 years old for repeat tympanostomy tube placement. At the time he was experiencing chronic fluid drainage from his ears that grew *S. aureus*. His physical examination revealed a prominent forehead and wide nasal bridge. Thick white rhinorrhea was present, and healing patches of eczema were noted on his skin. Given Stephen's history of multiple infections, an immunologist was consulted. A complete blood count revealed $1040 \mu\text{L}^{-1}$ white blood cells (normal $1049\text{--}5970 \mu\text{L}^{-1}$), of which 42% were neutrophils (normal 32–75%), 46% lymphocytes (normal 11–54%), and 7% eosinophils (normal 1–4%). Serum IgG was 1417 mg dL^{-1} (normal $600\text{--}1500 \text{ mg dL}^{-1}$), IgA was 70 mg dL^{-1} (normal $50\text{--}150 \text{ mg dL}^{-1}$), IgM was elevated at 145 mg dL^{-1} (normal $22\text{--}100 \text{ mg dL}^{-1}$), and IgE was markedly elevated at $36,698 \text{ mg dL}^{-1}$ (normal $0\text{--}200 \text{ mg dL}^{-1}$).

Sequencing of Stephen's *STAT3* gene revealed a heterozygous missense mutation in the highly conserved DNA-binding domain. CD4⁺ T cells were isolated from Stephen's peripheral blood, and intracellular staining for IL-17A, a marker of T_H17 lineage differentiation, was assessed by flow cytometry, revealing significantly decreased IL-17A expression compared with a healthy control (Fig. 20.3). Stephen was started on intravenous antibiotics after repeat tympanostomy tube placement. He was discharged to complete a course of antibiotics for his otitis media. He was then started on Bactrim prophylaxis, which led to a marked decrease in the frequency of infections.

Eosinophilia, high serum IgE.

Heterozygous dominant-negative mutation in *STAT3*.

Severe eczema, staphylococcal boils, recurrent sinopulmonary infections.

Hyper IgE syndrome.

Hyper IgE syndrome (HIES) is a rare primary immunodeficiency characterized by a triad of symptoms—recurrent staphylococcal skin abscesses, pneumonia, and very high serum IgE levels. It is also called Job's syndrome, in reference to the Biblical figure Job, who was "smote with sore boils." It was first described in 1966 by S. D. Davis and Ralph Wedgwood in two red-headed sisters with recurrent staphylococcal



Fig. 20.4 Chest radiograph of a patient with hyper IgE syndrome and STAT3 deficiency, showing a pneumatocele in the right upper lobe of the lung. (From VCU Health System Pediatric Radiology (<http://www.pedsradiology.com/>) with permission from Dr. Lakshmana D. Narla.)

skin abscesses that lacked typical features of inflammation such as redness and warmth, thus coining the term “cold abscesses.” The disease was further characterized a few years later by R. Buckley and colleagues, who noted the association between these cold abscesses, severe dermatitis, and high serum IgE levels and proposed the name “hyper IgE syndrome.” Patients with HIES are extremely susceptible to infections with *S. aureus* and the yeast *Candida albicans* and typically do not mount strong inflammatory responses such as fever to infections.

Several immunologic and non-immunologic phenomena have been described in HIES. An eczematous rash in the neonatal period is typically the first clinical manifestation. Staphylococcal skin abscesses occur in the majority of patients and begin early in life. Recurrent pneumonias occur and are often complicated by the formation of pneumatoceles (thin-walled air-filled cysts in the lungs that can result from scarring) (Fig. 20.4) and dilation of the large airways. The bacteria causing pneumonia include *S. aureus*, *Streptococcus pneumoniae*, and *Haemophilus influenzae*. Systemic signs of illness such as fever are often absent. Mucocutaneous candidiasis is common. A typical facial appearance develops during childhood, characterized by facial asymmetry, a broad nose, deep-set eyes, and a prominent forehead. Musculoskeletal abnormalities are common and include scoliosis (curvature of the spine), fractures occurring in the setting of minimal trauma, and hyperextensibility of the joints. Most patients have delayed shedding of their primary teeth, requiring surgical extraction (Fig. 20.5). There is an increased incidence of malignancies, including lymphoma and leukemia.

HIES is a clinical diagnosis based on a constellation of clinical and laboratory findings. A scoring system was developed by the National Institutes of Health, and patients with scores greater than 40 are more likely to have HIES. Both autosomal dominant and autosomal recessive forms of HIES have been described. In all cases the final common pathway is impaired T_H17 differentiation. T_H17 cells are important for maintaining mucosal and epithelial surface immunity against extracellular fungi and bacteria. They recruit neutrophils and macrophages to sites of infection by inducing epithelial cells expressing the IL-17 receptor to produce chemotactic factors for neutrophils (such as the chemokine CXCL8) and macrophages. In

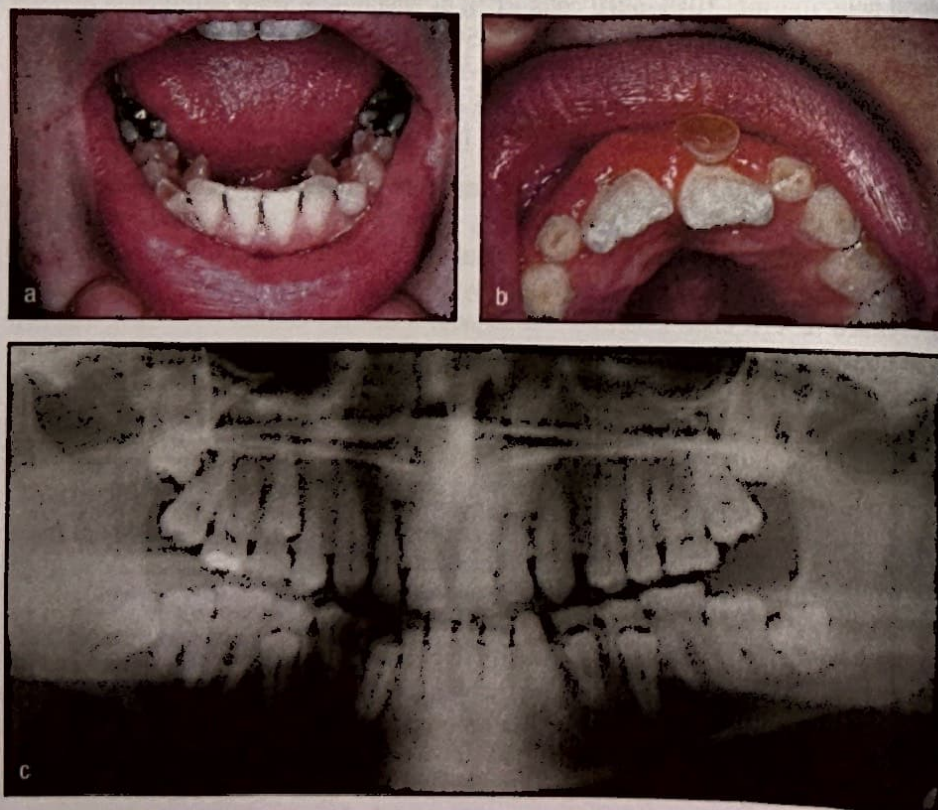


Fig. 20.5 Abnormal dentition in patients with hyper IgE syndrome and STAT3 deficiency.

In addition, T_H17 cells produce IL-22, which promotes the synthesis by keratinocytes of β -defensins, which are important in skin immunity. The mechanism by which the impaired T_H17 differentiation leads to elevated serum IgE is not yet understood.

The autosomal dominant form of HIES is the most common and has both sporadic and familial inheritance. It is a multisystem disease, and its features include skeletal and dental anomalies as well as recurrent bacterial infections. Heterozygous mutations in the gene for the transcription factor STAT3, located on chromosome 17, have been identified in the majority of patients with autosomal dominant HIES, resulting in a dominant-negative effect, as in Stephen. Most mutations occur within the DNA-binding, Src-homology (SH2), and transactivation domains of STAT3.

Autosomal recessive HIES is not typically associated with skeletal anomalies. In addition to bacterial infections, patients develop recurrent and severe infections of skin and mucous membranes with herpes simplex virus and the molluscum contagiosum virus, and are prone to develop squamous cell carcinomas. Mutations in the genes encoding dedicator of cytokinesis 8 (DOCK8) have been implicated in autosomal recessive HIES (see Case 50). Homozygous mutations in phosphoglucomutase 3 (PGM3) have recently been associated with HIES. Initially Tyk2 deficiency was reported as a cause of HIES in one patient. It is now clear that all patients with Tyk2 deficiency, including the one with HIES, suffer from susceptibility to mycobacterial and viral infections but do not have HIES. The cause for HIES in the initially reported case with Tyk2 deficiency is not known.

Mouse models of STAT3 deficiency have been developed. The homozygous mouse STAT3 knockout was embryonic lethal, possibly explaining why no humans who are homozygous for STAT3 deficiency have been identified. Mice with tissue-specific STAT3 deletions have been developed and show poor wound healing, eosinophilia, impaired β -defensin production, and pulmonary cyst formation, corresponding to some of the non-immunologic phenotypes seen in humans with HIES.

Questions.

- 1 Would a bone marrow transplant be useful in correcting the immunologic defects seen in HIES?
- 2 Some authors have found that patients with HIES due to STAT3 mutation had a fourfold decrease in expression of ROR γ T. Explain how this could be possible.
- 3 Administration of interferon-gamma (IFN- γ) has been used in HIES in an attempt to decrease serum IgE. The rationale is that IFN- γ counteracts IgE class switching induced by cytokines produced by T_H2 cells. Only a mild decrease in serum IgE was observed in some trials, and no reduction in others. What could be the reason for this?
- 4 Can you think of possible ways in which STAT3 deficiency could cause high levels of IgE?